

MS&E 228: Applied Causal Inference Powered by ML and AI

Lecture 14: What Can We Do About Unobserved Confounding?

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Winter 2026

Readings: *Applied Causal Inference Powered by ML and AI*, §12.1-12.2. Bellemare, Bloem, and Wexler (2024).

Goals of this Lecture

What can we do when we have unobserved confounding?

1. **The Problem:** What can we do when we have identified an unobserved confounder in our DAG?
2. **Quantifying Bias:** How can we quantify the bias from omitting an unobserved confounder?
3. **Sensitivity Analysis:** How can we use sensitivity analysis to bound the true treatment effect?
4. **Beyond Conditioning:** Are there identification strategies beyond conditioning that work even with unobserved confounding?

Where We Are: The Unobserved Confounder Problem

We have a powerful toolkit for causal inference:

- ▶ We can build DAGs from domain knowledge.
- ▶ We can use SWIGs to formally check for conditional exogeneity.
- ▶ We can identify valid adjustment sets (good controls) and avoid bad controls.

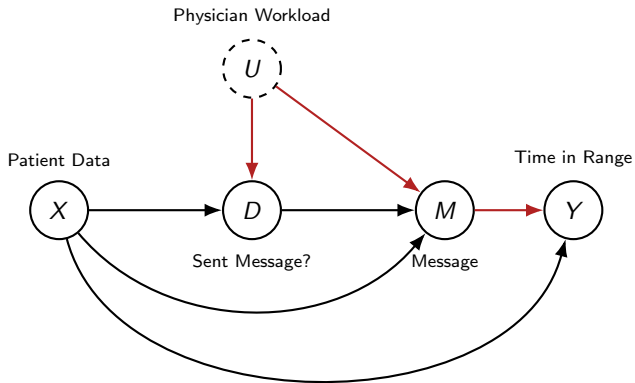
The Critical Failure Point

But what if the SWIG reveals an unobserved confounder that we cannot adjust for? For instance, the path $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$ in our hospital example was left open because physician workload (U_c) was unobserved.

Conditional exogeneity **fails**. Should we just give up?

Running Example: The Hospital Setting

Let's keep our running example in mind.



The Problem: Physician workload (U) is unobserved. It likely affects both the decision to send a message (D) and the patient's outcome (Y).

Poll Question

Poll: What to do first?

You have built a DAG for your application and have concluded that there is a key unobserved confounder that you have not measured. What is the first thing you would do?

What Are Our Options When Exogeneity Fails?

When faced with an unobserved confounder, we have a few paths forward:

1. **Find a Proxy:** Go back to the data owners and see if a good proxy for the unobserved confounder has been measured. If so, we can use it as a control.
2. **Sensitivity Analysis:** Give up on a point estimate and instead provide **bounds** on the treatment effect, based on assumptions about how strong the unobserved confounding is.
3. **Alternative Identification Strategies:** Use a different causal identification formula that does not rely on observing all of the confounders.

In this lecture, we will explore options 2 and 3.

Part I: Sensitivity Analysis

Bounding the Effect Under Unobserved Confounding

A Simple Case: Partially Linear Model with a Confounder

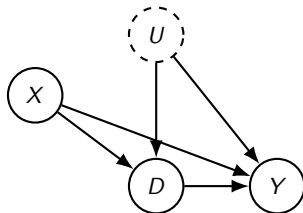
Let's start with a simple case to build intuition. Suppose we have one unobserved confounder U and relationships are partially linear.

Structural Equations:

$$Y = \theta D + \delta U + f_Y(X) + \varepsilon_Y$$

$$D = \gamma U + f_D(X) + \varepsilon_D$$

- ▶ U is an unobserved confounder (e.g., physician workload).
- ▶ We normalize $E[U^2] = 1$.
- ▶ We want to estimate the true causal effect θ .



Problem: We cannot condition on U . What happens if we run DoubleML just on X ?

What Does DoubleML Estimate Here?

Recall that DoubleML on controls X converges to the population OLS coefficient on the residuals:

$$\theta_s = \frac{E[\tilde{Y}\tilde{D}]}{E[\tilde{D}^2]}$$

where $\tilde{Y} = Y - E[Y|X]$ and $\tilde{D} = D - E[D|X]$.

Let's residualize our structural equations. Residualization is a linear operation ($U \perp\!\!\!\perp X$):

$$\tilde{Y} = \theta\tilde{D} + \delta U + \varepsilon_Y$$

$$\tilde{D} = \gamma U + \varepsilon_D$$

So, running DML with controls X is equivalent to running OLS on this simpler residualized system of equations.

Deriving the Omitted Variable Bias (1/2)

What is the OLS estimand θ_s in this residualized system?

$$\theta_s = \frac{E[\tilde{Y}\tilde{D}]}{E[\tilde{D}^2]} = \frac{E[(\theta\tilde{D} + \delta U + \varepsilon_Y)\tilde{D}]}{E[\tilde{D}^2]}$$

Since $E[\varepsilon_Y\tilde{D}] = 0$ (by construction of ε_Y):

$$\theta_s = \theta + \delta \frac{E[U\tilde{D}]}{E[\tilde{D}^2]}$$

Now we need to figure out what $E[U\tilde{D}]$ is.

Deriving the Omitted Variable Bias (2/2)

Let's expand $E[U\tilde{D}]$ using the residualized equation for $\tilde{D}$:

$$\tilde{D} = \gamma U + \varepsilon_D$$

So,

$$E[U\tilde{D}] = E[U(\gamma U + \varepsilon_D)] = E[\gamma U^2 + U\varepsilon_D] = \gamma E[U^2] + E[U\varepsilon_D]$$

Since U is a confounder, it is independent of the idiosyncratic error ε_D , so $E[U\varepsilon_D] = 0$. And since we normalized $E[U^2] = 1$, we get:

$$E[U\tilde{D}] = \gamma$$

The Omitted Variable Bias Formula (Simple Case)

Plugging $E[U\tilde{D}] = \gamma$ back into our expression for θ_s :

$$\theta_s = \theta + \delta \frac{\gamma}{E[\tilde{D}^2]}$$

Omitted Variable Bias Formula (Simple Case)

This gives us the famous omitted variable bias formula:

$$\text{Bias} = |\theta_s - \theta| = \frac{|\delta\gamma|}{E[\tilde{D}^2]}$$

Interpreting the Bias Formula

The bias depends on three quantities:

$$|\theta_s - \theta| = \frac{|\delta| \cdot |\gamma|}{E[\tilde{D}^2]}$$

$|\delta|$: How much the unobserved confounder U influences the outcome Y .

$|\gamma|$: How much the unobserved confounder U influences the treatment D .

$E[\tilde{D}^2]$: The variance of the treatment residual after controlling for X . This is **measurable from data**.

The Path to Sensitivity Analysis

We can estimate θ_s and $E[\tilde{D}^2]$ from data. If we are willing to make **assumptions** on the strength of confounding (posit bounds on $|\delta|$ and $|\gamma|$), we can compute bounds on the true effect θ .

Part II: General Sensitivity Analysis

From Linear Coefficients to Explained Variance

Generalizing the Analysis

This was a bit simplified. What if we have a more general, non-linear model?

General NP-SEM with Unobserved Confounding:

$$Y = f_Y(D, X, U, \varepsilon_Y)$$

$$D = f_D(X, U, \varepsilon_D)$$

All we assume is that if we had observed U , the model would be identified via DoubleML. This holds if the CEF is partially linear in D after conditioning on X, U :

$$E[Y|D, X, U] = \theta D + g(X, U)$$

This is the same assumption that justifies DML, just with U included in the controls.

If we run DML omitting U , we again converge to a wrong quantity θ_s . Can we characterize the bias?

Measuring Confounding Strength via Explained Variance

Instead of linear coefficients δ and γ , we use a more general measure: **partial R-squared**.

This measures the *relative gain in explained variance* from adding U to a model.

Strength of Confounder on Outcome: $\eta_{Y \sim U|D,X}^2$

Instead of δ^2 , we use the partial R^2 of U on Y , after controlling for D, X .

Definition: Partial R^2 of U on Y

$$\eta_{Y \sim U|D,X}^2 = \frac{E[\text{Var}(Y|D, X)] - E[\text{Var}(Y|D, X, U)]}{E[\text{Var}(Y|D, X)]}$$

Interpretation of Partial R^2 of U on Y

$$\eta_{Y \sim U|D,X}^2 = 1 - \frac{\text{Unexplained variance of } Y \text{ from } D, X, U}{\text{Unexplained variance of } Y \text{ from } D, X}$$

The relative gain in explained variance of Y by adding U to a model that already has D, X .

Strength of Confounder on Treatment: $\eta_{D \sim U|X}^2$

Instead of γ^2 , we use the partial R^2 of U on D , after controlling for X .

Definition: Partial R^2 of U on D

$$\eta_{D \sim U|X}^2 = \frac{\mathbb{E}[\text{Var}(D|X)] - \mathbb{E}[\text{Var}(D|X, U)]}{\mathbb{E}[\text{Var}(D|X)]}$$

Interpretation of Partial R^2 of U on D

$$\eta_{D \sim U|X}^2 = 1 - \frac{\text{Unexplained variance of } D \text{ from } X, U}{\text{Unexplained variance of } D \text{ from } X}$$

The relative gain in explained variance of D by adding U to a model that already has X .

The General Omitted Variable Bias Formula

With these more general measures, the bias formula is remarkably similar.

Omitted Variable Bias Formula (General Case)

$$|\theta_s - \theta|^2 = \eta_{Y \sim U|D,X}^2 \cdot \frac{\eta_{D \sim U|X}^2}{1 - \eta_{D \sim U|X}^2} \cdot \frac{\mathbb{E}[\text{Var}(Y|D, X)]}{\mathbb{E}[\text{Var}(D|X)]}$$

The terms $E[\text{Var}(Y|D, X)]$ and $E[\text{Var}(D|X)]$ are estimable from the DML residuals:

- ▶ $\mathbb{E}[\text{Var}(Y|D, X)] = \mathbb{E}[(Y - \mathbb{E}[Y|D, X])^2] = E[(\tilde{Y} - \hat{\theta}_s \tilde{D})^2]$
- ▶ $\mathbb{E}[\text{Var}(D|X)] = \mathbb{E}[(D - \mathbb{E}[D|X])^2] = \mathbb{E}[\tilde{D}^2]$

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$$|\theta_s - \theta|^2 = \eta_{Y \sim U|D, X}^2 \cdot \frac{\eta_{D \sim U|X}^2}{1 - \eta_{D \sim U|X}^2} \cdot \frac{\mathbb{E}[\text{Var}(Y|D, X)]}{\mathbb{E}[\text{Var}(D|X)]}$$

The Path to Sensitivity Analysis

Once again, if we can posit bounds on the confounding strength (η^2 terms), we can bound the true effect θ , since everything else is measurable from data.

Footnote: Pattern Matching the Formulas

How does the general formula relate to the simple linear case?

$$|\theta_s - \theta|^2 = \frac{\delta^2 \gamma^2}{(\mathbb{E}[\tilde{D}^2])^2}$$

We can show that the terms correspond:

- ▶ $\delta^2 = \frac{\eta_{Y \sim U|D,X}^2}{1 - \eta_{D \sim U|X}^2} \cdot \mathbb{E}[\text{Var}(Y|D, X)]$
- ▶ $\gamma^2 = \eta_{D \sim U|X}^2 \cdot \mathbb{E}[\text{Var}(D|X)] = \eta_{D \sim U|X}^2 \cdot \mathbb{E}[\tilde{D}^2]$

This provides a mental model for how the more complex partial R^2 terms map back to the intuitive linear coefficients.

Part III: The Sensitivity Analysis Procedure

From Theory to Practice

The Sensitivity Analysis Procedure (1/2)

1. Run DML with observed controls X to get an estimate $\hat{\theta}_s$ and residuals $\check{Y}, \check{D}$.
2. Estimate the data-dependent scaling factor:

$$\hat{S} = \sqrt{\frac{E_n[(\check{Y} - \hat{\theta}_s \check{D})^2]}{E_n[\check{D}^2]}}$$

The Sensitivity Analysis Procedure (2/2)

3. Ask a domain expert to provide plausible upper bounds on confounding strength:
 - ▶ η_Y^2 : At most, how much **more** variance of the outcome could U explain?
 - ▶ η_D^2 : At most, how much **more** variance of the treatment could U explain?
4. Calculate the bounds on the true effect θ :

$$\hat{\theta}_s \pm \sqrt{\frac{\eta_Y^2 \cdot \eta_D^2}{1 - \eta_D^2}} \cdot \hat{S}$$

Confidence Bounds

We can also produce a bound that holds with 95% confidence by incorporating the standard error of our estimates.

Confidence Bounds for θ :

$$\left[\hat{\theta}_{lower} - 1.96 \cdot SE_{lower}, \quad \hat{\theta}_{upper} + 1.96 \cdot SE_{upper} \right]$$

where $\hat{\theta}_{lower}$ and $\hat{\theta}_{upper}$ are the bounds from the previous slide.

The standard errors must account for the uncertainty in estimating $\hat{\theta}_s$ and $\hat{S}$. Using techniques similar to the DML in-sensitivity theorem the standard errors for the lower and upper bound can be easily calculated.

Pseudocode for Sensitivity Bounds

```
# res = (y_residuals, D_residuals) from DML
def dml_sensitivity_bounds(res, eta_ysq, eta_dsq):
    y_res, D_res = res
    nu_sq = np.mean(D_res ** 2)
    theta_s = np.mean(y_res * D_res) / nu_sq
    sigma_sq = np.mean((y_res - D_res * theta_s)**2)

    S = np.sqrt(sigma_sq / nu_sq)
    C_d_sq = eta_dsq / (1 - eta_dsq)
    C_g_sq = eta_ysq

    error = S * np.sqrt(C_d_sq * C_g_sq)

    return theta_s - error, theta_s + error
```

Click on [this notebook](#) for a full implementation on the Darfur conflict data.

Part IV: Making Sensitivity Practical

Benchmarking and the Robustness Value

Benchmarking and Contour Plots

Coming up with plausible bounds for η_Y^2 and η_D^2 can be hard. What can we do?

Idea: Benchmarking against observed covariates.

1. For each observed control X_j (or group of controls), calculate how much explained variance it adds to the outcome and treatment on top of the other controls.
2. This gives us a concrete sense of scale for the η^2 values.

Contour Plots

We can visualize the bounds as a function of η_Y^2 and η_D^2 using contour plots. We can then plot the benchmark values from observed covariates on these plots.

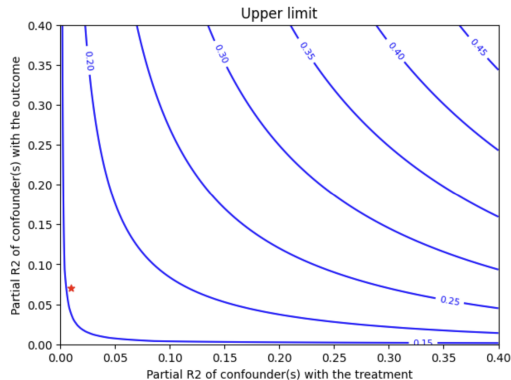
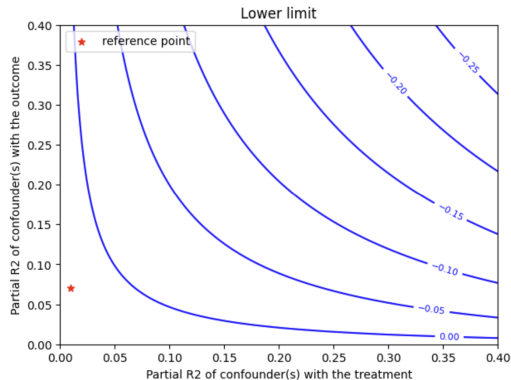
This allows us to ask questions like: *"If our unobserved confounder is twice as strong as 'age', would our conclusion still hold?"*

Benchmarking and Contour Plots in the Darfour Example

Estimate effect of prior war exposure (D) on attitude towards peace (Y), controlling for demographics X (sex, age, farmer, herder, past-voted, household size).

Double ML effect ≈ 0.10

All observed controls explain $\approx 7\%$ of outcome and $\lesssim 1\%$ of treatment variance.



The Robustness Value

A very useful summary of the sensitivity analysis is the **robustness value**.

Definition: Robustness Value (RV)

The robustness value is the value c^2 such that if the unobserved confounder explained c^2 of the residual variance in both the treatment and outcome (i.e., $\eta_Y^2 = \eta_D^2 = c^2$), the 95% confidence interval for the treatment effect would just include zero.

Interpretation: It tells you how strong an unobserved confounder would need to be to nullify your statistical finding.

Example: If your RV is 5%, you can say: "An unobserved confounder would need to explain at least 5% of the residual variance in both treatment and outcome to change our conclusion that the effect is non-zero."

Part V: Beyond Partial Linearity

Sensitivity for Binary Treatments

What if the Effect is Not Constant?

The partial linearity assumption $E[Y|D, X, U] = \theta D + g(X, U)$ implies a **constant** treatment effect θ .

This is a strong assumption, especially for binary treatments where personalized effects are likely.

Example: Hospital Setting

In our hospital example, we don't want to assume the effect of sending a message is the same for every patient. The benefit of a message likely depends on patient characteristics (X).

We need a sensitivity analysis framework that allows for **heterogeneous treatment effects**.

Sensitivity for Binary Treatments

For binary treatments, we can relax the constant effect assumption. The bias characterization is very similar, but uses a slightly different measure for the confounder's effect on the treatment.

- ▶ The measure of confounding on the outcome, $\eta_{Y \sim U|D,X}^2$, remains the same.
- ▶ The measure of confounding on the treatment, $\eta_{D \sim U|X}^2$, changes.

Instead of gain in explained variance, we use gain in **precision** (inverse variance).

New Confounding Measure for Treatment

Definition: Gain in Precision for Treatment

$$\eta_{D \sim U|X}^2 = \frac{E[1/\text{Var}(D|X, U)] - E[1/\text{Var}(D|X)]}{E[1/\text{Var}(D|X, U)]} \in [0, 1]$$

Interpretation: How much more precise our prediction of the treatment D becomes when we add the unobserved confounder U to our model.

Bias Formula for Binary Treatments

With this new measure, the bias formula is analogous.

If we use the Doubly Robust estimator (which allows for HTE) instead of DML, we converge to a biased estimate θ_s such that:

$$|\theta_s - \theta|^2 = \eta_{Y \sim U|D,X}^2 \cdot \frac{\eta_{D \sim U|X}^2}{1 - \eta_{D \sim U|X}^2} \cdot S^2$$

where the scaling factor S^2 is now:

$$S^2 = E[\text{Var}(Y|D, X)] \cdot E[1/\text{Var}(D|X)]$$

and can also be estimated from data.

Applying to the Hospital Example

How would we reason about the bounds in our running example?

- ▶ **Bound on η_Y^2 :** How much variance in patient outcomes (time in range) could physician workload explain, on top of patient data and whether they got a message? *Probably small.*
- ▶ **Bound on η_D^2 :** How much more precise would our prediction of 'message sent' be if we knew the physician's workload? *It would be a factor, but likely not a substantial one, as the decision is primarily driven by patient monitoring data.*

Part VI: Beyond Conditioning

The Front-Door Criterion

Back to Basics: What is Identification?

We have been so focused on achieving identification by finding a valid **adjustment set** to satisfy conditional exogeneity.

Omitted variable bias and sensitivity analysis are direct consequences of this approach failing.

But let's step back. What is identification?

Identification

Identification is the exercise of expressing a causal quantity of interest (e.g., ATE) as a statistical quantity that can be measured from observational data.

Key insight: Adjusting for confounders is **not the only way** to achieve identification. Other formulas exist that rely on different causal structures.

The Front-Door Criterion: An Example

Let's consider an example from Bellemare, Bloem, and Wexler (2024) on ride-sharing.

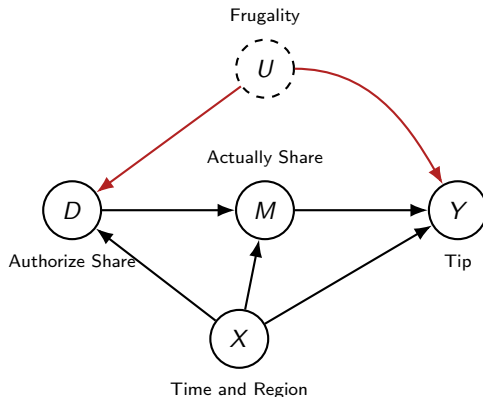
Uber/Lyft Tipping Example

- ▶ ***D***: Passenger **authorizes** a shared ride (e.g., chooses UberX Share).
- ▶ ***M***: Passenger **actually shares** the ride with another person.
- ▶ ***X***: time-of-day, source region and target region.
- ▶ ***Y***: Tipping behavior (e.g., tip percentage).

The problem: Frugal people are more likely to authorize sharing ($U \rightarrow D$) and also more likely to tip less ($U \rightarrow Y$). We have unobserved confounding.

The Front-Door DAG

This scenario corresponds to a specific causal graph structure.



The confounding path $D \leftarrow U \rightarrow Y$ is open.
We cannot use the adjustment formula.

The Front-Door Conditions

For the front-door criterion to hold, we need a special structure:

1. There is a variable (or set of variables) M that intercepts all directed paths from D to Y . (i.e., $D \rightarrow M \rightarrow Y$)
2. There is no unblocked back-door path from D to M , after conditioning on X . (i.e., D is as-if random with respect to M , conditional on X)
3. All back-door paths from M to Y are blocked by D, X .

In our DAG, M (Actually Share) satisfies these conditions.

How the Front-Door Criterion Works (1/2)

We can identify the total effect of D on Y by decomposing it into two parts that *are* identified.

Step 1: Identify the effect of D on M

$$\gamma = \text{Effect}(D \rightarrow M) = \mathbb{E}[\mathbb{E}[M \mid D = 1, X] - \mathbb{E}[M \mid D = 0, X]]$$

Why? Because the relationship is **unconfounded**. There is no open back-door path from D to M , once conditioning on X . We can simply regress M on D, X .

How the Front-Door Criterion Works (2/2)

Step 2: Identify the effect of M on Y

$$\delta = \text{Effect}(M \rightarrow Y) = \mathbb{E}[\mathbb{E}[Y \mid M = 1, D, X] - \mathbb{E}[Y \mid M = 0, D, X]]$$

Why? Because all back-door paths from M to Y (e.g. $M \leftarrow D \leftarrow U \rightarrow Y$) are **blocked by conditioning on D, X** .

The Front-Door Formula

Effect of D on Y is the product of the effect of D on M times the effect of M on Y

$$\theta = \gamma \cdot \delta$$

So simply for linear models:

1. Run $\text{OLS}(M \sim D, X)$, let $\hat{\gamma}$ be coefficient on D
2. Run $\text{OLS}(Y \sim M, D, X)$, let $\hat{\delta}$ be coefficient on M
3. Return $\hat{\theta} = \hat{\gamma} \cdot \hat{\delta}$

Requires an extra subtle assumption that individual effect of D on M is independent of individual effect of M on Y .

Otherwise, "product of average effects" is not equal to average of "product of effects"

The Front-Door Formula

More generally, we can estimate effect of D on the whole distribution of M and effect of M on the whole distribution of Y and then combine, to get the effect of D on the distribution of Y :

The Front-Door Formula

$$P(Y=y|\text{do}(D=d)) = \sum_m P(M=m|D=d) \sum_{d'} P(Y=y|M=m, D=d')P(D=d')$$

Intuition: The causal effect is a weighted combination of the effect of $D \rightarrow M$ and the effect of $M \rightarrow Y$.

Note: The simplistic idea of just multiplying the effects only works in linear models without interaction.

Poll Question

Poll: Front-Door Identification

In the ride-sharing example, why can we identify the effect of **actually sharing** (M) on **tipping** (Y)?

1. Because D is a pre-treatment variable.
2. Because conditioning on D blocks the back-door path from M to Y .
3. Because the relationship between M and Y is unconfounded by nature.
4. Because we assume a linear relationship.

The Broader Point: Many Identification Strategies Exist

The Front-Door criterion is just one example of an identification strategy that does not rely on observing all confounders.

In fact, given any causal DAG, there is a complete algorithm, called the **ID algorithm**, that can determine if a causal effect is identifiable. If it is, the algorithm returns the correct statistical formula to compute it.

The ID Algorithm

The **ID algorithm** (e.g., implemented in packages like DoWhy in Python) is a black-box algorithm that takes as input the DAG and automatically find an identification formula if one exists.

What if the Effect is Not Identifiable?

What happens if the ID algorithm fails and tells you the effect is not identifiable from the graph alone?

This means you cannot identify the effect without making **further assumptions** beyond the parental relationships in the DAG.

Examples of further assumptions:

- ▶ Linearity of structural equations.
- ▶ Monotonicity assumptions.
- ▶ Re-interpreting the causal quantity you are after.

Many techniques in econometrics rely on such auxiliary assumptions.

Next Time...

Instrumental Variables

A powerful and widely used identification strategy for unobserved confounding that relies on finding a special variable (the instrument) with specific properties.

Takeaways

1. **Sensitivity analysis bounds the effect:** When unobserved confounding is present, we can't get a point estimate, but we can provide bounds on the true effect by making assumptions on the confounding strength.
2. **Bias is intuitive:** The omitted variable bias is characterized by gains in explained variance (the η^2 quantities), which provides an intuitive way for domain experts to reason about confounding.
3. **Benchmarking provides scale:** To make sensitivity analysis practical, we can benchmark the required confounding strength against the strength of observed covariates and use contour plots to visualize robustness.
4. **Identification is more than just adjustment:** The front-door criterion shows that other causal structures allow for identification even with unobserved confounding. Always consider the full DAG to see if alternative strategies apply.

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